

Project 3 – Microsoft Intune Endpoint Management Lab."

Project Overview

This project demonstrates how Microsoft Intune can be used to manage and secure Windows devices within a cloud-based environment. The lab covers the complete device lifecycle, including enrollment, compliance management, configuration profiles, application deployment, update management, endpoint security, device retirement, and Windows Autopilot.

The objective of this project was to gain hands-on experience with Microsoft Intune and understand how organisations can centrally manage endpoints, enforce security policies, automate software deployment, and maintain device compliance through a modern cloud management platform.

Business Scenario

A retail organisation is expanding its use of cloud-based endpoint management to support staff working across multiple store locations. Instead of manually configuring each Windows device, the organisation needs a centralised solution to enroll devices, apply security policies, deploy required applications, manage updates, and protect company data.

In this lab, I used Microsoft Intune to simulate how an IT support team can manage Windows endpoints for retail staff. The focus was on improving device security, reducing manual setup work, and ensuring that company devices remain compliant with organisational requirements.

Environment and Tools Used

- Microsoft Intune Admin Center
- Microsoft Entra ID
- Microsoft 365 Developer Tenant
- Windows 11 Virtual Machine
- Microsoft Defender Antivirus
- Windows Autopilot
- Microsoft 365 Apps for Enterprise

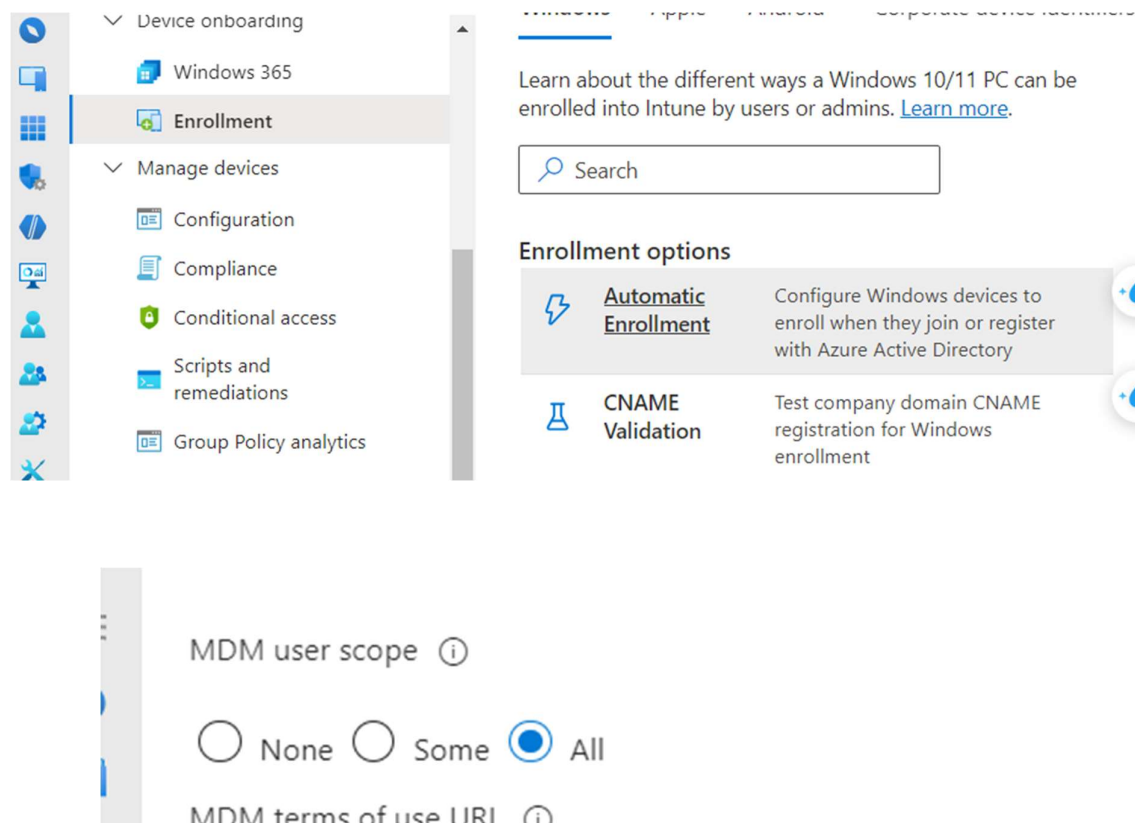
Phase 1- Device Enrollment

Objective

To configure Microsoft Intune enrollment settings and successfully onboard a Windows 11 device for centralized management, monitoring, and policy deployment.

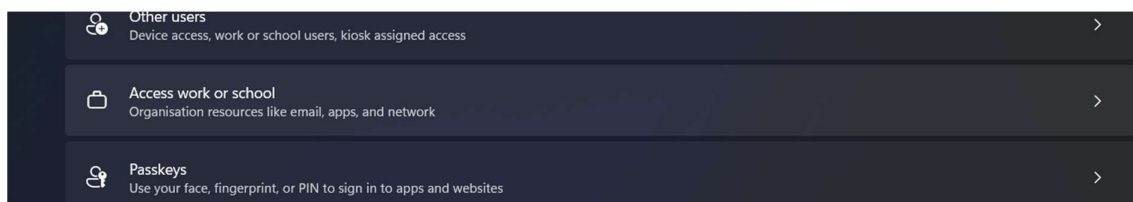
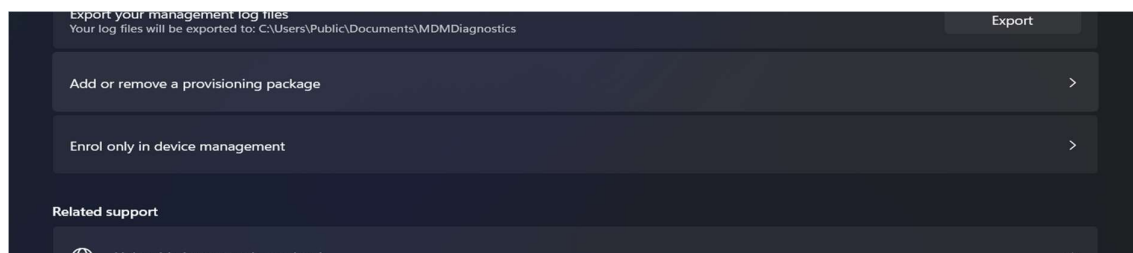
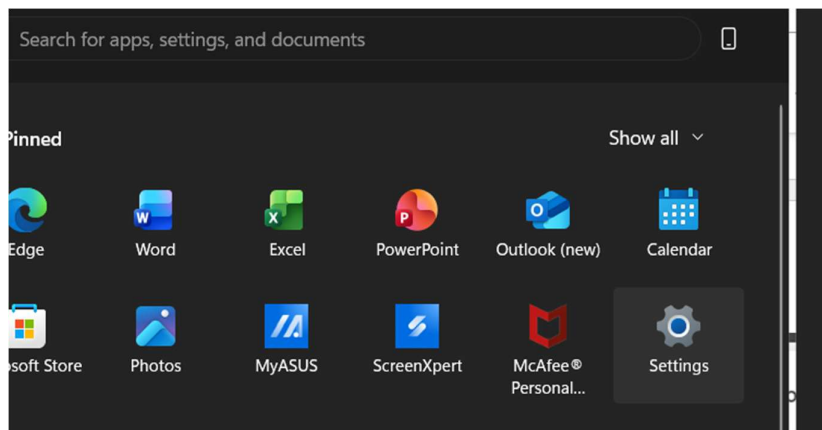
Step 1 – Configure Automatic Enrollment

To begin this phase, I configured **Automatic Enrollment** within Microsoft Intune by enabling the **MDM User Scope** for all users. This allows licensed users to automatically enroll their Windows devices into Intune when signing in with their organisational account.



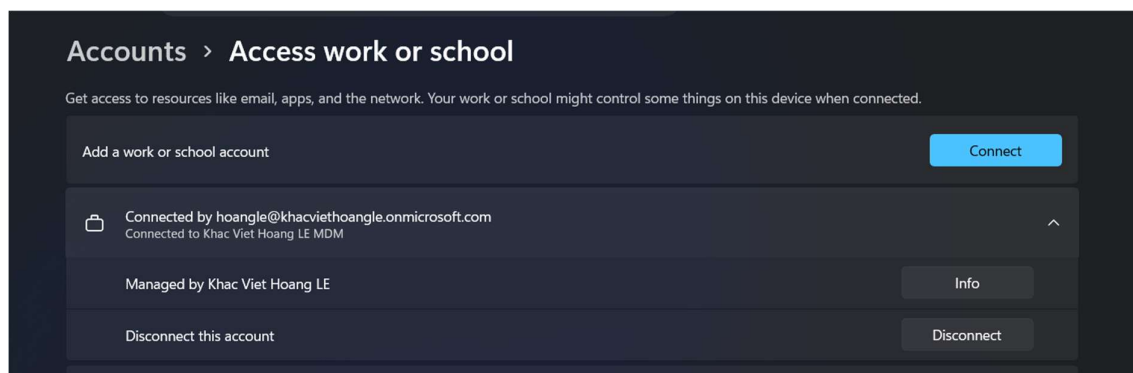
Step 2 – Enroll Windows Device into Intune

After configuring enrollment settings, I enrolled a Windows 11 device by navigating to **Access work or school** and selecting **Enroll only in device management**. This method allowed the device to be managed by Intune without performing a full Microsoft Entra ID join.



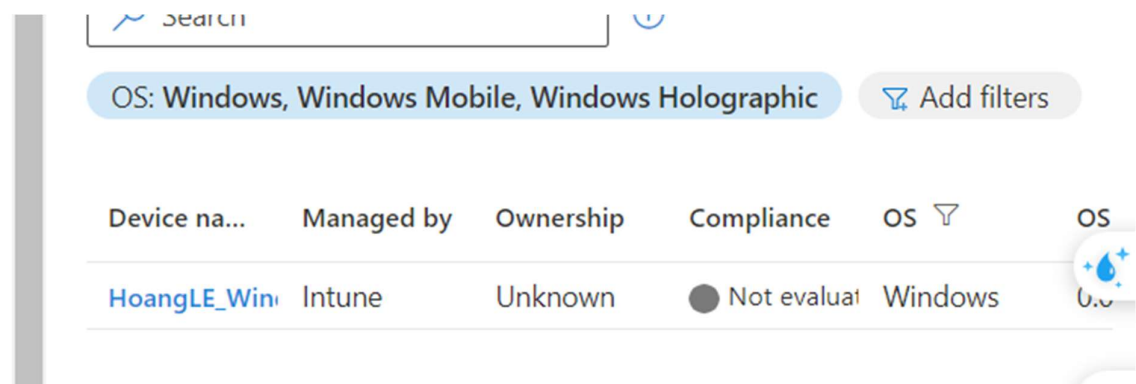
Step 3 – Verify Device Enrollment

Once the enrollment process was completed, the device successfully connected to the organisation's MDM environment. This confirmed that Intune was now able to manage the endpoint remotely.



After signing in with my Microsoft 365 organisational account, the device successfully enrolled into Intune and appeared in the **Windows Devices** management portal.

Initially, the device displayed a **“Not Evaluated”** compliance status, which is expected because no compliance policies had been assigned yet at this stage.



Search					
OS: Windows, Windows Mobile, Windows Holographic				Add filters	
Device na...	Managed by	Ownership	Compliance	OS	OS
HoangLE_Win	Intune	Unknown	Not evaluat	Windows	Windows

Once enrolled, the device became available for remote management, allowing administrators to apply policies, monitor compliance, deploy applications, and manage endpoint configurations centrally.

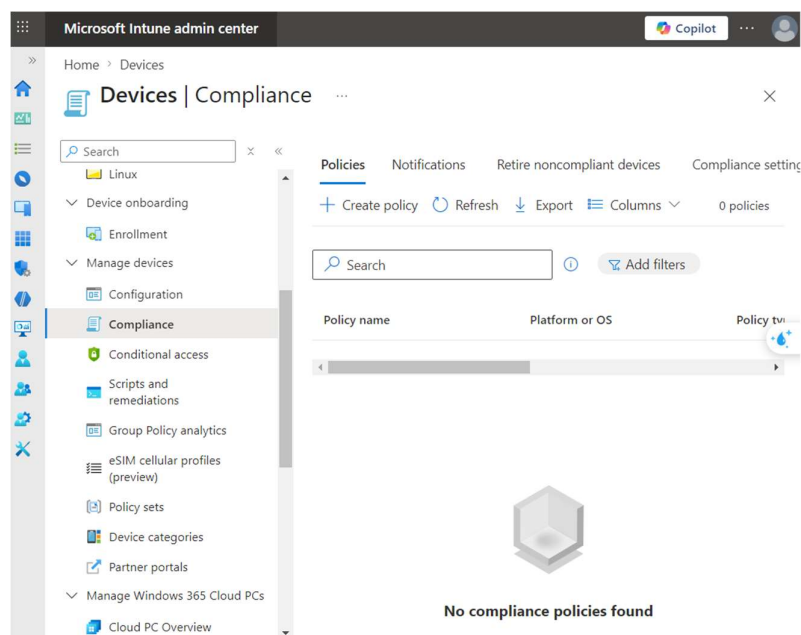
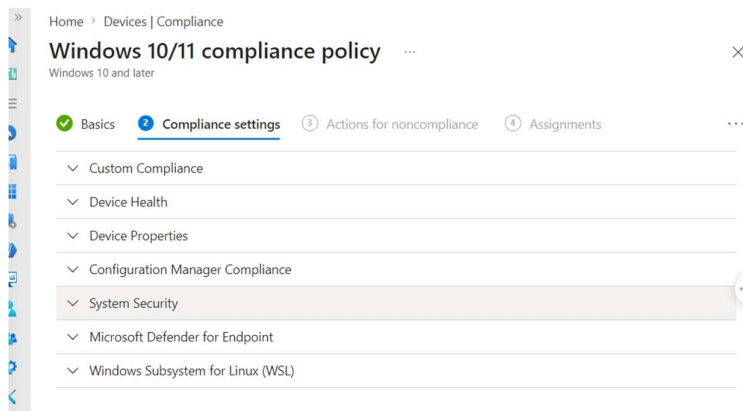
Phase 2- Compliance Policy (Require PIN / Password - Require Encryption (BitLocker))

Objective

Create a compliance policy to ensure Windows devices meet the organisation's minimum security requirements before accessing company resources.

Step 1 – Create Compliance Policy

To strengthen endpoint security, I created a Windows compliance policy in Microsoft Intune. This policy was designed to evaluate whether managed devices met the organisation's minimum security requirements before being considered compliant.



Step 2 – Configure Compliance Requirements

Within the compliance settings, I configured **password protection** and **device encryption requirements**. Devices were required to use a password or PIN and have encryption enabled to help protect organisational data from unauthorised access, especially if a device was lost or stolen.

^ System Security

Password

Require a password to unlock mobile devices ⓘ

RequireNot configured

Simple passwords ⓘ

BlockNot configured

Password type ⓘ

Device default

Minimum password length ⓘ

6✓

Maximum minutes of inactivity before password is required ⓘ

Not configured

Password expiration (days) ⓘ

41

Number of previous passwords to prevent reuse ⓘ

5

Require password when device returns from idle state (Mobile and Holographic) ⓘ

RequireNot configured

I also configured the noncompliance action to immediately mark devices as noncompliant if they failed to meet the required security standards. This allows administrators to quickly identify devices that may pose a security risk.

✓ Basics ✓ Compliance settings 3 Actions for noncompliance 4 Assignments

Specify the sequence of actions on noncompliant devices

Action	Schedule (days after noncompliance) ⓘ	Message template	Additional recipients (...)
Mark device noncompliant	Immediately		
Add device to retire list	Immediately		...

▼

0

Step 3 – Assign and Deploy Compliance Policy

Finally, I assigned the compliance policy to the **Retail_Store_Staff group** and reviewed the configuration before deployment. Once deployed, the policy automatically evaluated enrolled devices against the defined security requirements.

✔ Compliance settings

✔ Actions for noncompliance

✔ Assignments

5 Review + create

...

Summary

Basics

Name

Windows Endpoint Compliance Policy

Description

This policy enforces security requirements including password/PIN and encryption settings for managed Windows devices.

Platform

Windows 10 and later

Profile type

Windows 10/11 compliance policy

Compliance settings

System Security

Require a password to unlock mobile devices

Require

Simple passwords

Block

Minimum password length

6

Require encryption of data storage on device.

Require

Firewall

Require

Trusted Platform Module (TPM)

Require

Actions for noncompliance

Action	Schedule	Message template	Additional recipients (via ...
Mark device noncompliant	Immediately		

Previous

Create

Result- Managed Windows devices were successfully evaluated against password and encryption requirements, helping ensure that only secure and compliant devices could access organisational resources.

Phase 3- Configuration Profile

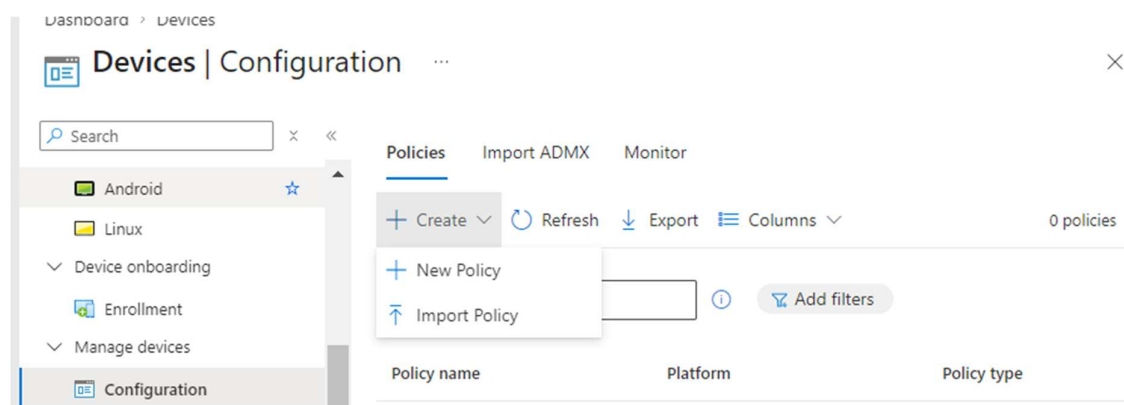
Part 1- Password Policy

Objective

Configure password and device lock settings to strengthen endpoint security and enforce consistent authentication requirements across managed Windows devices.

Step 1 - Create Configuration Profile

To improve endpoint security, I created a configuration profile in Microsoft Intune using the Settings Catalog. This profile was designed to enforce password and device lock requirements across managed Windows devices.



Step 2 -Settings Catalog / Device Lock

Using the Settings Catalog, I selected the Device Lock category to configure password-related controls. This allowed me to centrally manage authentication and lock screen settings for enrolled devices.

1 Basics 2 Configuration settings 3 Scope tags 4 Assignments 5 Review + create



Settings catalog

With the settings catalog, you can choose which settings you want to configure. Click on Add settings to browse or search the catalog for the settings you want to configure.

[Learn more](#)

[+ Add settings](#)

14 settings in "Device Lock" category

Select all these settings

Setting name	
☐ [Deprecated] Max Inactivity Time Device Lock With External Display	ⓘ
☒ Account Lockout Policy	ⓘ
> ☒ Device Password Enabled	ⓘ
☐ Enforce Lock Screen And Logon Image	ⓘ
☐ Enforce Lock Screen Provider	
☒ Minimum Password Age	ⓘ

Step 3 - Password Configuration

I configured several password security settings, including a minimum password length of six characters, password history enforcement, account lockout protection, and automatic device locking after 15 minutes of inactivity. These settings help reduce the risk of weak passwords, unauthorised access, and brute-force attacks.

^ Device Lock

Remove category

ⓘ 13 of 14 settings in this category are not configured

Account Lockout Policy ⓘ ✓ ⊖

^ Device Lock

Remove category

ⓘ 5 of 14 settings in this category are not configured

Device Password Enabled ⓘ ☒ Enabled ⊖

Allow Simple Device Password ⓘ ⊖

Alphanumeric Device Password Required ⓘ ⊖

Device Password Expiration * ⓘ ✓ ⊖

Device Password History * ⓘ ✓ ⊖

Max Device Password Failed Attempts * ⓘ ✓ ⊖

Max Inactivity Time Device Lock * ⓘ ✓ ⊖

Min Device Password Length * ⓘ ✓ ⊖

Account Lockout Policy ⓘ ✓ ⊖

Step 4 - Review + Create

After reviewing the configuration, I assigned the profile to the **Retail_Store_Staff** group and deployed the policy. Once applied, all targeted devices automatically received the password and device lock settings defined in the profile.

Progress: ✓ Basics ✓ Configuration settings ✓ Scope tags 4 Assignments 5 Review + create

Included groups

[Add groups](#) [Add all users](#) [Add all devices](#)

Groups	Status	Group Members ⓘ	Filter	Filter mode
No groups selected				

Excluded groups

[+ Add groups](#)

ⓘ When excluding groups, you cannot mix user and device groups across include and exclude. [Click here to learn more about excluding groups.](#)

Groups	Status	Group Members ⓘ	Remove
Retail_Store_Staff	Active	0 devices, 4 users	Remove

Create profile

Windows 10 and later - Settings catalog

Progress: ✓ Basics ✓ Configuration settings ✓ Scope tags ✓ Assignments 4 Review + create

Summary

Basics

Name	Windows Security Configuration Profile
Description	Enforces password policy and device restrictions on managed Windows endpoints.
Platform	Windows

Configuration settings

Device Lock

Account Lockout Policy ⓘ	5
Device Password Enabled ⓘ	Enabled
Allow Simple Device Password ⓘ	Not allowed.
Alphanumeric Device Password Required ⓘ	Password, Numeric PIN, or Alphanumeric PIN required.
Device Password Expiration ⓘ	30
Device Password History ⓘ	5
Max Device Password Failed Attempts ⓘ	5

[Previous](#) [Create](#)

Result- A standardised password policy was successfully deployed to managed Windows devices, helping improve authentication security and ensure consistent endpoint protection across the organisation.

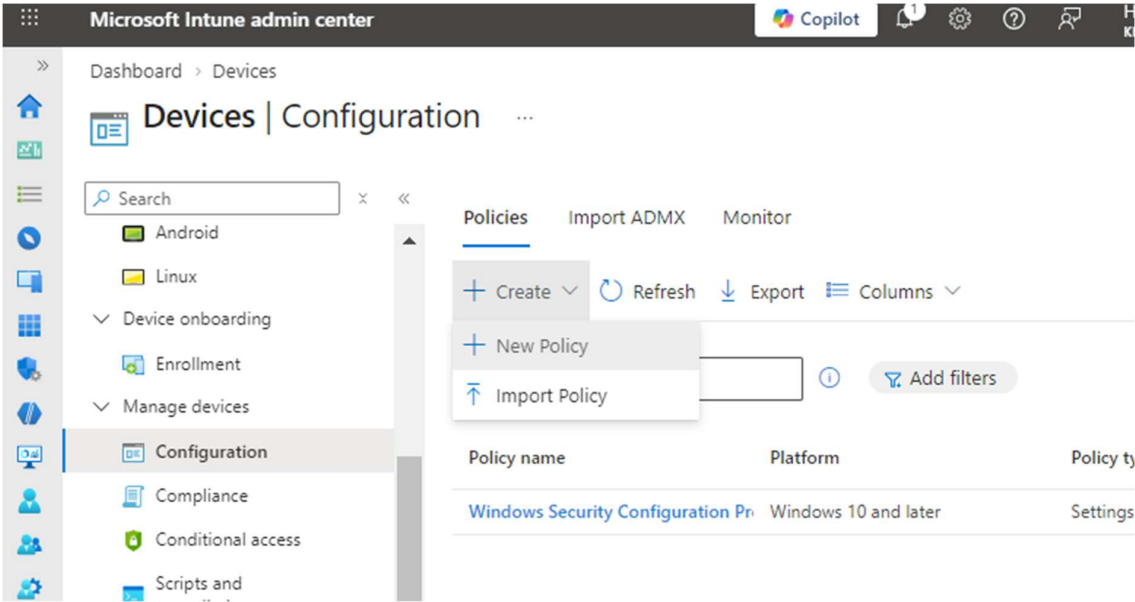
Part 2.Device Restrictions

Objective

Configure device restriction settings to limit access to sensitive system features and reduce the risk of unauthorised changes on managed Windows devices.

Step 1- Create Device Restrictions Profile

After configuring password policies, I created a separate configuration profile to apply additional device restriction settings. This profile was designed to prevent users from making unauthorised changes to company-managed devices.



Step 2- Control Panel Restriction

Within the **Administrative Templates > Control Panel** category, I enabled **Prohibit access to Control Panel and PC settings (User)**.

By enabling this setting, users are prevented from accessing the Windows Control Panel and Settings application. This helps reduce accidental misconfigurations and ensures that only authorised administrators can modify important system settings.

Dashboard > D

reate pr

dows 10 and la

Basics

Add settings (

Administ

Control

5 of 6

Prohibit a

settings (l

Settings picker

Use commas "," among search terms to lookup settings by their keywords

control panel

Search

+ Add filter

Browse by category

Administrative Templates\Control Panel

Administrative Templates\Control Panel\Add or Remove Programs

Administrative Templates\Control Panel\Display

Administrative Templates\Control Panel\Personalization

Administrative Templates\Control Panel\Printers

Administrative Templates\Control Panel\Programs

Administrative Templates\Control Panel\Regional and Language Options

Administrative Templates\Control Panel\Regional and Language Options\Handwriting personalization

Administrative Templates\Control Panel\User Accounts

Administrative Templates\Start Menu and Taskbar

6 results in the "Control Panel" subcategory

Select all these settings

Setting name

☐ Always open All Control Panel Items when opening Control Panel (User)

☐ Hide specified Control Panel items (User)

☒ Prohibit access to Control Panel and PC settings (User)

☐ Show only specified Control Panel items (User)

Administrative Templates

Remove category

Control Panel

Remove subcategory

5 of 6 settings in this subcategory are not configured

Prohibit access to Control Panel and PC settings (User)

Enabled

Step 3 - System Restore Restrictions

I also disabled **System Restore** and related recovery **configuration** options. This prevents users from rolling back device settings or restoring previous system states that could bypass organisational policies and security controls.

Create profile

Windows 10 and later - Settings catalog

✓ Basics

2 Configuration settings

③ Scope tags

④ Assignments

⑤ Review + create

+ Add settings ⓘ

^ Administrative Templates

Remove category

System > System Restore

Remove subcategory

Turn off Configuration ⓘ

Enabled

⊖

Turn off System Restore ⓘ

Enabled

⊖

Control Panel

Remove subcategory

ⓘ 5 of 6 settings in this subcategory are not configured

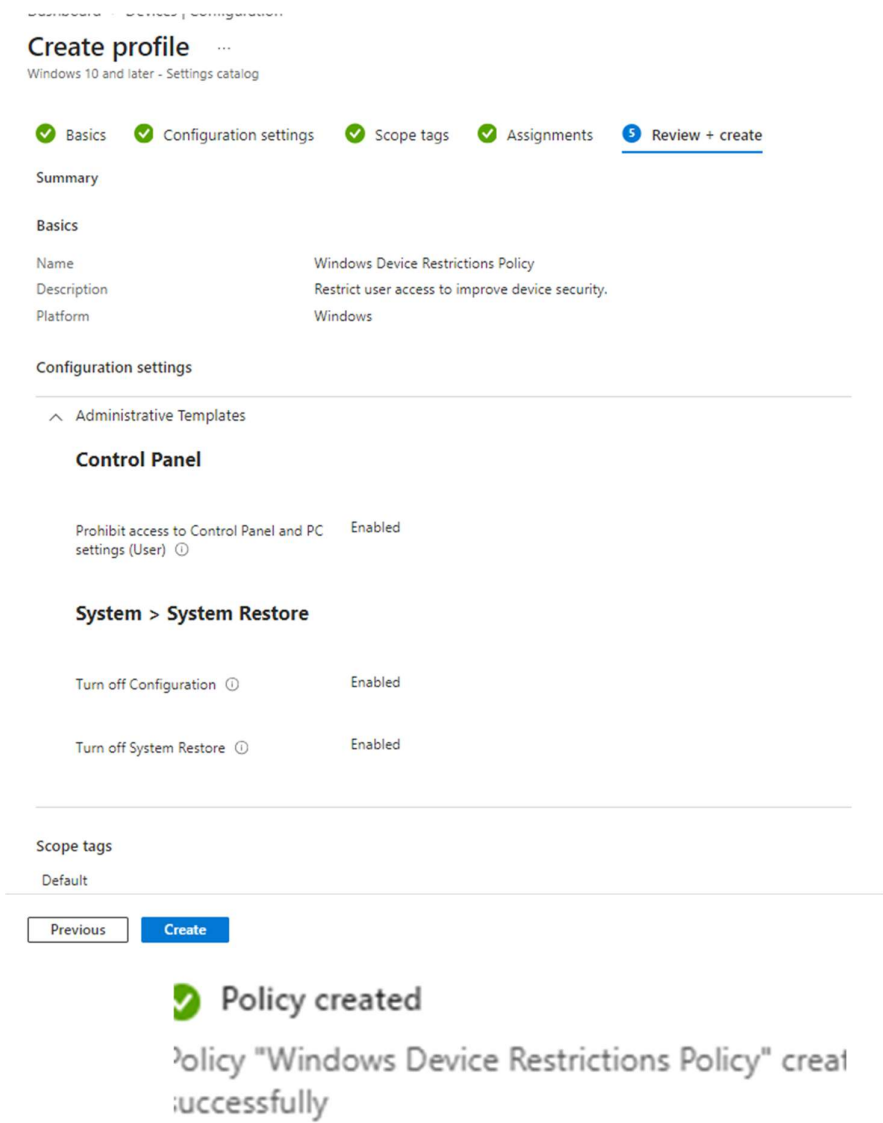
Prohibit access to Control Panel and PC settings (User) ⓘ

Enabled

⊖

Step 4 - Review + Create

After reviewing the configuration, I assigned the profile to the **Retail_Store_Staff** group and deployed the policy. Once applied, the restriction settings were automatically enforced across all targeted devices.



Result- Device restriction settings were successfully deployed, reducing the ability for users to modify critical system settings and helping maintain a consistent and secure endpoint environment.

Phase 4 – Application Deployment

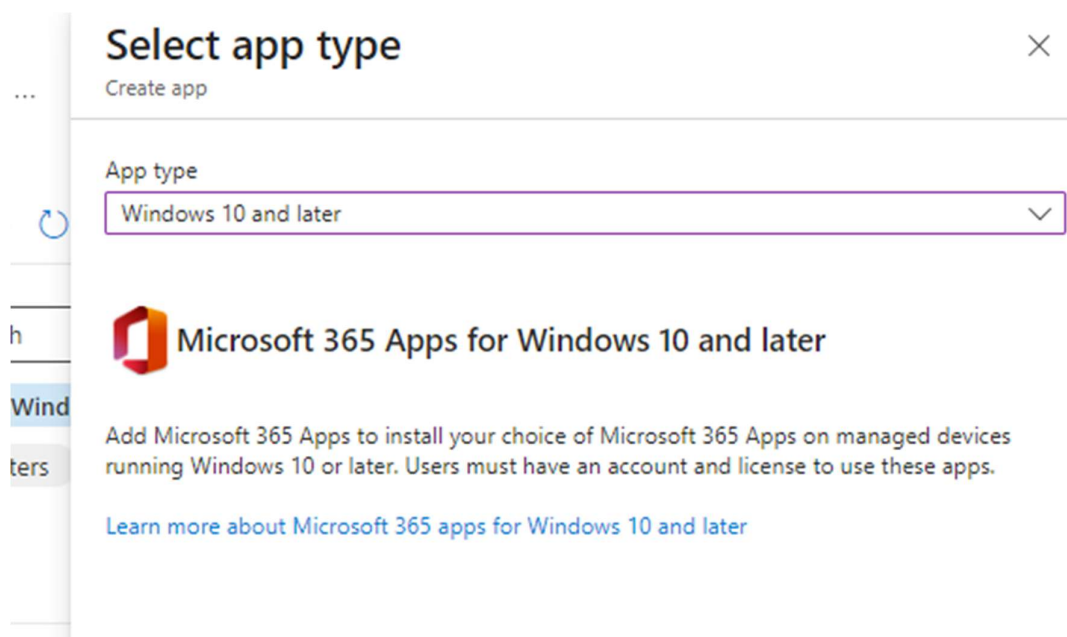
Part 1. Required deployment (auto installation)

Objective

Deploy essential productivity applications automatically to managed devices using Microsoft Intune.

Step 1 - Create Microsoft 365 Application

To streamline software deployment, I created a Microsoft 365 application package using **Microsoft 365 Apps for Windows 10 and later**. This deployment method allows Microsoft 365 applications to be installed directly on managed Windows devices without requiring separate installation files.



Step 2 - Configure Microsoft 365 Apps

Within the application suite configuration, I selected **Word, Excel, Outlook, PowerPoint, and Teams** as the standard productivity applications for staff devices. These applications were chosen because they are commonly used for communication, document management, and daily business operations.

Dashboard > Apps | Overview

Add Microsoft 365 Apps

Microsoft 365 Apps (Windows 10 and later)

Select Office apps ⓘ

Select other Office apps (license required) ⓘ

App suite information

These settings apply to all apps you have selected

Architecture ⓘ

Default file format *

Update channel * ⓘ

Remove other versions ⓘ

Version to install ⓘ

Specific version

Properties

Use shared computer activation ⓘ

Accept the Microsoft Software License Terms on behalf of users

Install background service for Microsoft Search in Bing ⓘ

Languages ⓘ

5 selected

- ☐ Access
- ☒ Excel
- ☐ OneNote
- ☒ Outlook
- ☒ PowerPoint
- ☐ Publisher
- ☐ Skype for Business
- ☒ Teams
- ☒ Word

Yes No

Latest Specific

Latest version

Yes No

Yes No

Yes No

No languages selected

Previous Next

Step 3 -Deployment Settings

I configured the deployment to use **64-bit architecture**, **Office Open XML Format**, and the **Monthly Enterprise Channel** for updates. I also enabled **Remove Other Versions** to automatically replace any existing Office installations, ensuring consistency across all managed devices.

App suite information

These settings apply to all apps you have selected in the suite. [Learn more](#)

Architecture ⓘ	☐ 32-bit ☒ 64-bit
Default file format *	Office Open Document Format
Update channel * ⓘ	Monthly Enterprise Channel
Remove other versions ⓘ	☒ Yes ☐ No
Version to install ⓘ	☒ Latest ☐ Specific
Specific version	Latest version

Properties

Use shared computer activation ⓘ	☐ Yes ☒ No
Accept the Microsoft Software License Terms on behalf of users	☒ Yes ☐ No
Install background service for Microsoft Search in Bing ⓘ	☒ Yes ☐ No
Languages ⓘ	No languages selected

Previous

Next

Step 4 - Required Assignment

After that, I moved to the **Assignments** section. Under **Required**, I assigned the application to the **Retail_Store_Staff** group. This means the selected Microsoft 365 applications will be installed automatically on devices assigned to users in that group, without requiring the user to manually start the installation.

✓ App suite information

✓ Configure app suite

3 Assignments

4 Review + create

Required ⓘ

Group mode	Group	Status	Filter mode	Filter	
+ Included	Retail_Store_Staff	Active	None	None	**

[+ Add group](#) ⓘ [+ Add all users](#) ⓘ [+ Add all devices](#) ⓘ

Available for enrolled devices ⓘ

Group mode	Group	Status	Filter mode	Filter
No assignments				

[+ Add group](#) ⓘ [+ Add all users](#) ⓘ

Uninstall ⓘ

Group mode	Group	Status	Filter mode	Filter
No assignments				

[+ Add group](#) ⓘ [+ Add all users](#) ⓘ [+ Add all devices](#) ⓘ

Finally, I reviewed the deployment settings and clicked **Create** to deploy the application successfully. This required deployment helps ensure that important productivity applications are installed consistently across organisational devices

✓ App suite information

✓ Configure app suite

✓ Assignments

4 Review + create

Summary

App suite information

Name

Microsoft 365 Required Deployment

Description

Required productivity applications for staff devices

Publisher

Microsoft

Category

Productivity

Show this as a featured app

No

Information URL

<https://products.office.com/explore-office-for-home>

Privacy URL

<https://privacy.microsoft.com/privacystatement>

Developer

Microsoft

Owner

Microsoft

Notes

No Notes

Logo



Configure app suite

Apps to be installed as part of the suite

Excel, Outlook, PowerPoint, Teams, Word

✓ Application created

Application "Microsoft 365 Required Deployment" created successfully

Result- Microsoft 365 applications were successfully deployed automatically to targeted devices through Microsoft Intune, ensuring consistent software availability across the organisation.

Part 2. Available Deployment (Self-Service Install)

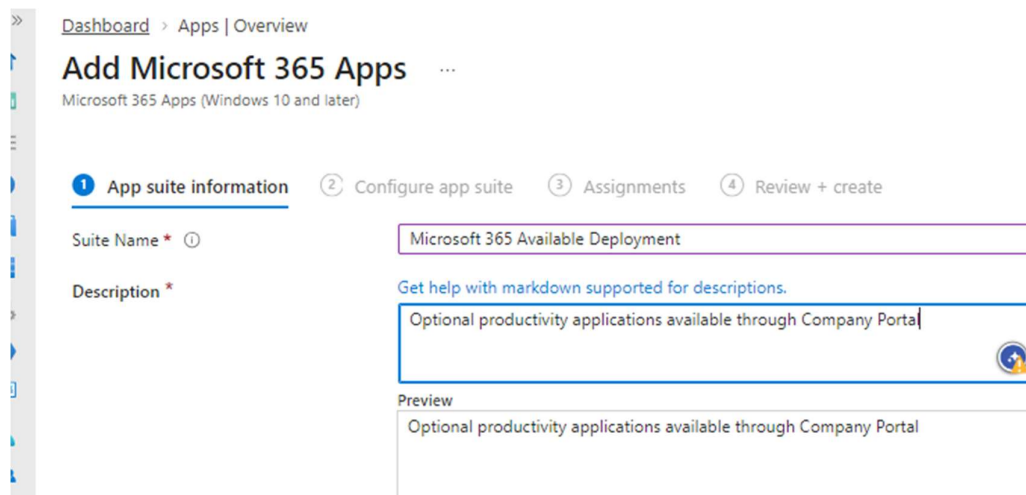
Objective

Provide users with the ability to install optional applications on demand through the Company Portal.

Step 1 - Available Deployment Profile

To provide users with greater flexibility, I created a second deployment profile named **Microsoft 365 Available Deployment**. Unlike the required deployment, this profile was intended for optional productivity applications that users could install when needed through the Company Portal.

The deployment included the standard Microsoft 365 application suite, including Word, Excel, Outlook, PowerPoint, and Teams, ensuring that users had access to approved business applications while maintaining organisational software standards.



Dashboard > Apps | Overview

Add Microsoft 365 Apps

Microsoft 365 Apps (Windows 10 and later)

1 App suite information 2 Configure app suite 3 Assignments 4 Review + create

Suite Name * ① Microsoft 365 Available Deployment

Description * Get help with markdown supported for descriptions.
Optional productivity applications available through Company Portal

Preview
Optional productivity applications available through Company Portal

Step 2 - Application Configuration and Assignment

Within the application configuration settings, I selected **64-bit Architecture**, **Office Open XML Format**, and the **Monthly Enterprise Channel** to maintain consistency with the required deployment profile. I also enabled **Remove Other Versions** to automatically replace older Office installations and reduce software compatibility issues.

Microsoft 365 Apps (Windows 10 and later)

1 App suite information 2 **Configure app suite** 3 Assignments 4 Review + create

Configuration settings format * Configuration designer

Configure app suite

Select Office apps ① 5 selected

Select other Office apps (license required) ① 0 selected

App suite information

These settings apply to all apps you have selected in the suite. [Learn more](#)

Architecture ① 32-bit 64-bit

Default file format * Office Open XML Format

Update channel * ① Monthly Enterprise Channel

Remove other versions ① Yes No

Version to install ① Latest Specific

Specific version Latest version

Properties

Use shared computer activation ① Yes No

Accept the Microsoft Software License Terms on behalf of users Yes No

Instead of assigning the application as **Required**, I deployed it under **Available for enrolled devices** and targeted the **Retail_Store_Staff** group. This allows authorised users to install the application manually through the Company Portal whenever required.

1 App suite information 2 Configure app suite 3 **Assignments** 4 Review + create

Required ①

Group mode	Group	Status	Filter mode	Filter
No assignments				

+ Add group ① + Add all users ① + Add all devices ①

Available for enrolled devices ①

Group mode	Group	Status	Filter mode	Filter
Included	Retail_Store_Staff	Active	None	None

+ Add group ① + Add all users ①

Uninstall ①

Group mode	Group	Status	Filter mode	Filter
No assignments				

Step 3 - Review and Deployment

Finally, I reviewed the deployment settings and selected **Create** to complete the application deployment. This deployment method provides users with flexibility by allowing optional applications to be installed on demand.

The screenshot displays the 'Review + create' step of the Microsoft 365 App Suite configuration process. The interface includes a progress bar at the top with four steps: 'App suite information', 'Configure app suite', 'Assignments', and 'Review + create' (the current step, indicated by a blue circle and number 4). Below the progress bar, the 'Summary' section is visible, followed by the 'App suite information' section. This section contains a table with the following details:

Name	Microsoft 365 Available Deployment
Description	Optional productivity applications available through Company Portal
Publisher	Microsoft
Category	Productivity
Show this as a featured app	No
Information URL	https://products.office.com/explore-office-for-home
Privacy URL	https://privacy.microsoft.com/privacystatement
Developer	Microsoft
Owner	Microsoft
Notes	No Notes
Logo	

Below the table, there is a 'Configure app suite' link. At the bottom of the screen, a green checkmark icon is followed by the text 'Application created', and a right-pointing chevron icon is visible. Below this, a message states: 'Application "Microsoft 365 Available Deployment" created successfully'.

Result- Microsoft 365 applications were successfully published through the Company Portal, allowing users to install approved software on demand while maintaining centralised application management and organisational security standards.

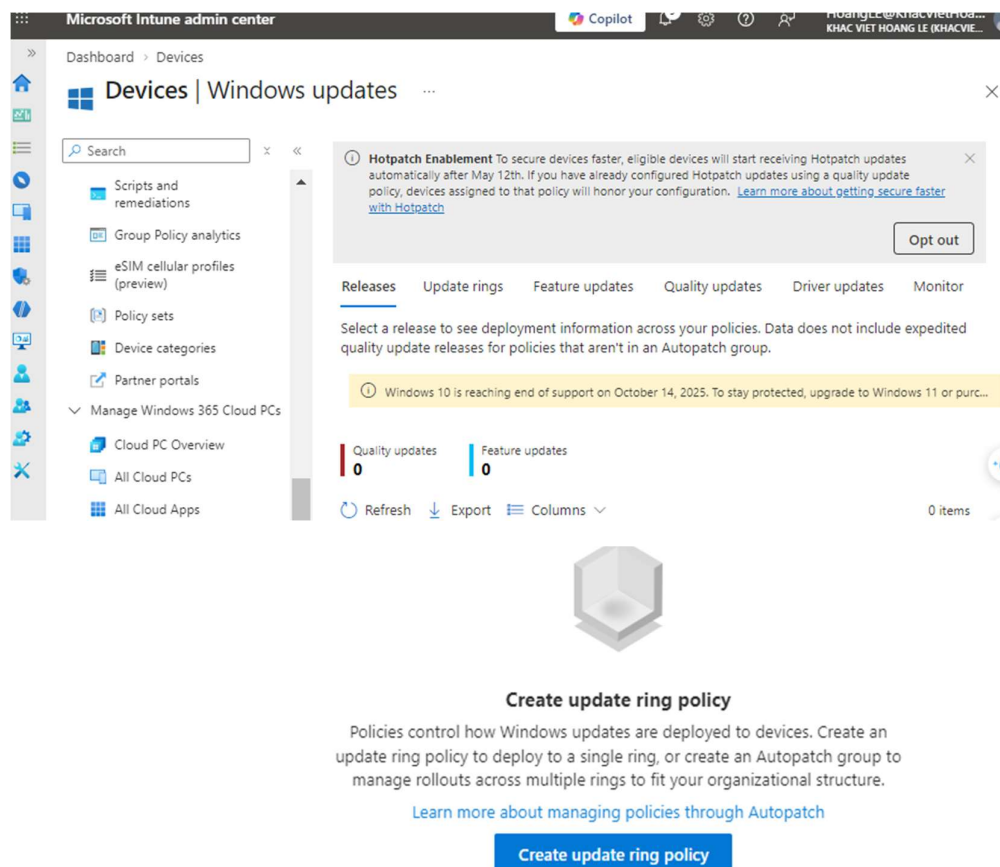
Phase 5 – Configure Windows Update Rings

Objective

Configure Windows Update for Business settings to automate update deployment, control restart behaviour, and maintain device security across managed endpoints.

Step 1- Create Update Ring Policy

To centrally manage Windows updates, I created a new update ring policy named **Windows Update Ring Policy** within Microsoft Intune. This policy was designed to automate update deployment and ensure that managed devices receive security updates in a controlled and consistent manner.



Step 2 – Configure Update Ring Settings

Within the update ring settings, I enabled **Microsoft Product Updates** and **Windows Drivers** to ensure that managed devices receive operating system, Microsoft application, and driver updates automatically.

I configured both **Quality Update Deferral Period** and **Feature Update Deferral Period** to **7 days**, allowing updates to be tested and stabilised before deployment. I also set **Upgrade Windows 10 devices to Latest Windows 11 release** to No to prevent

automatic operating system upgrades that could affect application compatibility or business operations.

The screenshot shows the 'Update ring settings' configuration page in the Microsoft Intune console. The page is divided into two main sections: 'Update settings' and 'User experience settings'. The 'Update settings' section includes options for Microsoft product updates, Windows drivers, quality update deferral period (7 days), feature update deferral period (7 days), upgrade Windows 10 devices to the latest Windows 11 release (No), set feature update uninstall period (10 days), enable pre-release builds (Not Configured), and select pre-release channel (Windows Insider - Release Preview). The 'User experience settings' section includes options for automatic update behavior (Auto install at maintenance time), active hours start (8 AM), active hours end (5 PM), option to pause Windows updates (Disable), and option to check for Windows updates (Enable).

1 Basics 2 Update ring settings 3 Assignments 4 Review + create

Update settings

Microsoft product updates * ⓘ ☒ Allow ☐ Block

Windows drivers * ⓘ ☒ Allow ☐ Block

Quality update deferral period (days) * ⓘ 7 ✓

Feature update deferral period (days) * ⓘ 7 ✓

Upgrade Windows 10 devices to latest Windows 11 release ⓘ ☐ Yes ☒ No

Set feature update uninstall period (2 - 60 days) * ⓘ 10

Enable pre-release builds * ⓘ ☐ Enable ☒ Not Configured

Select pre-release channel Windows Insider - Release Preview

User experience settings

Automatic update behavior ⓘ Auto install at maintenance time

Active hours start * ⓘ 8 AM

Active hours end * ⓘ 5 PM

Option to pause Windows updates ⓘ ☐ Enable ☒ Disable

Option to check for Windows updates ⓘ ☒ Enable ☐ Disable

Step 3- User Experience Settings and Assignment

For the **Automatic Update Behaviour**, I selected **Auto Install at Maintenance Time** so updates would install automatically during scheduled maintenance periods. I then configured **Active Hours Start** to **8 AM** and **Active Hours End** to **5 PM** so devices would avoid restarting during standard working hours.

Additionally, I set **Option to Pause Windows Updates** to **Disable** so users cannot postpone critical security updates, while leaving **Option to Check for Windows Updates** enabled to allow manual update checks when required group would receive the configured Windows update settings automatically.

User experience settings

Automatic update behavior ⓘ

Active hours start * ⓘ

Active hours end * ⓘ

Option to pause Windows updates ⓘ

Option to check for Windows updates ⓘ

Change notification update level ⓘ

Use deadline settings ⓘ

Deadline for feature updates ⓘ

Deadline for quality updates ⓘ

Grace period ⓘ

Auto reboot before deadline ⓘ

Step 4 – Review and Deployment

After reviewing the policy configuration, I deployed the update ring through Microsoft Intune. The policy was successfully created and assigned, allowing update management to be centrally controlled without requiring manual intervention on individual devices.

☒ Basics
 ☒ Update ring settings
 ☒ Assignments
 ☒ Review + create

Summary

Basics

Name Windows Update Ring Policy

Description Manage Windows update schedule and restart behaviour for company devices

Update ring settings

Update settings

Microsoft product updates	Allow
Windows drivers	Allow
Quality update deferral period (days)	7
Feature update deferral period (days)	7
Upgrade Windows 10 devices to Latest Windows 11 release	No
Set feature update uninstall period (2 - 60 days)	10
Servicing channel	General Availability channel

User experience settings

Automatic update behavior	Auto install at maintenance time
Active hours start	8 AM
Active hours end	5 PM
Option to pause Windows updates	Disable
Option to check for Windows updates	Enable
Change notification update level	Use the default Windows Update notifications
Use deadline settings	Not configured

Assignments

- ☒ **Update Ring successfully assigned**

Go to Windows Update Ring Policy to view the details.
- ☒ **Update Ring successfully created**

Go to Windows Update Ring Policy to view the details.

Result- Windows Update for Business settings were successfully deployed through Intune, enabling automated update installation, controlled restart behaviour, and consistent patch management across managed Windows devices.

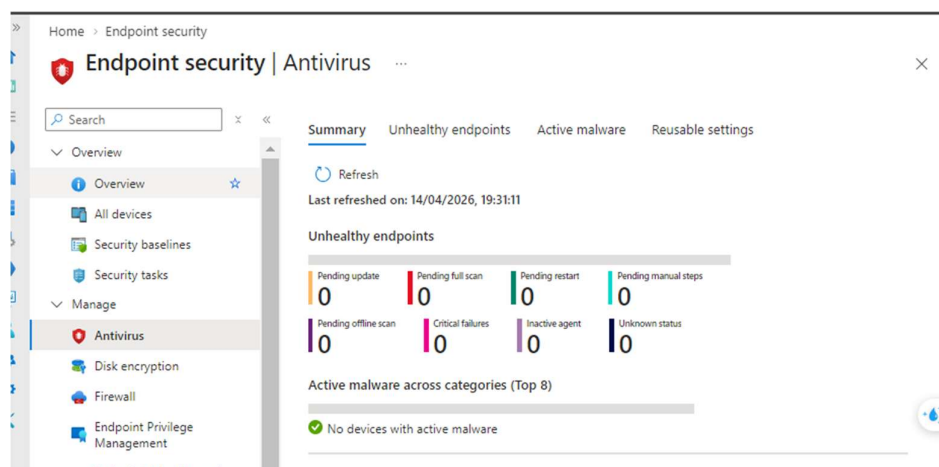
Phase 6 – Configure Endpoint Security Policies

Objective

Configure Microsoft Defender Antivirus policies through Microsoft Intune to strengthen endpoint protection against malware, malicious scripts, phishing attacks, and other security threats.

Step 1 – Create Microsoft Defender Antivirus Policy

To begin this Phase, I navigated to Endpoint security within the Microsoft Intune Admin Center and selected Antivirus. From there, I clicked Create Policy to begin configuring a new antivirus security policy for managed devices.



For the platform, I selected **Windows**, and for the profile, I chose **Microsoft Defender Antivirus**. I selected this profile because Microsoft Defender Antivirus is the built-in endpoint protection solution for Windows devices and is commonly used to manage antivirus security settings in an organisational environment.

Create a profile

Platform

Windows

Profile

Microsoft Defender Antivirus

Microsoft Defender Antivirus

Windows Defender Antivirus is the next-generation protection component of Microsoft Defender for Endpoint. Next-generation protection brings together machine learning, big-data analysis, in-depth threat resistance research, and cloud infrastructure to protect devices in your enterprise organization.

This policy applies to: Windows 10, Windows 11, and Windows Server

The settings in this policy can be targeted to: MDM, MicrosoftSense supported devices

Step 2 – Configure Defender Security Settings

Next, in the **Configuration Settings** section, I configured several important Microsoft Defender security controls. I enabled **Allow Archive Scanning** so compressed files such as ZIP files can also be scanned for threats. I enabled **Allow Behavior Monitoring** so Defender can monitor suspicious activity and detect malicious behaviour in real time. I also enabled **Allow Cloud Protection** so the device can use Microsoft cloud-based intelligence to improve threat detection.

Basics

Configuration settings

Scope tags

Assignments

Review + create

Search

Defender

Allow Archive Scanning

Allowed. Scans the archive files. (Default)

Allow Behavior Monitoring

Allowed. Turns on real-time behavior monitoring. (Default)

Allow Cloud Protection

Allowed. Turns on Cloud Protection. (Default)

Allow Email Scanning

Allowed. Turns on email scanning.

Allow Full Scan On Mapped Network Drives

Allowed. Scans mapped network drives.

Allow Full Scan Removable Drive Scanning

Allowed. Scans removable drives.

[Deprecated] Allow Intrusion Prevention System

Allowed. (Default)

Allow scanning of all downloaded files and attachments

Allowed. (Default)

Allow Realtime Monitoring

Allowed. Turns on and runs the real-time monitoring service. (Default)

Allow Scanning Network Files

Allowed. Scans network files. (Default)

Allow Script Scanning

Allowed. (Default)

In addition, I enabled **Allow Realtime Monitoring** so Defender can continuously scan files and processes while the device is in use. I enabled **Allow Script Scanning** to ensure that potentially harmful scripts can also be detected and blocked. I also enabled **Check for Signatures Before Running Scan** so the antivirus engine updates its signatures before performing a scan, helping improve detection accuracy

Step 3 – Network Protection Configuration

To further strengthen protection, I enabled **Check for Signatures Before Running Scan** to ensure the latest threat intelligence is downloaded before antivirus scans are performed. I also configured **Enable Network Protection** to **Block Mode**, allowing Defender to actively block access to malicious websites, phishing pages, and dangerous network-based content.

These controls provide an additional layer of protection beyond traditional malware scanning and help reduce the risk of web-based attacks.

Allow Realtime Monitoring ⓘ	Allowed. Turns on and runs the real-time monitoring service. (Default) ▾
Allow Scanning Network Files ⓘ	Allowed. Scans network files. (Default) ▾
Allow Script Scanning ⓘ	Allowed. (Default) ▾
Allow User UI Access ⓘ	Allowed. Lets users access UI. (Default) ▾
Avg CPU Load Factor ⓘ	☒ Not Configured
Archive Max Depth ⓘ	☒ Not Configured
Archive Max Size ⓘ	☒ Not Configured
Check For Signatures Before Running Scan ⓘ	Enabled ▾
Cloud Block Level ⓘ	Not configured ▾
Cloud Extended Timeout ⓘ	☒ Not Configured
Days To Retain Cleaned Malware ⓘ	☒ Not Configured
Disable Catchup Full Scan ⓘ	Not configured ▾
Disable Catchup Quick Scan ⓘ	Not configured ▾
Enable Low CPU Priority ⓘ	Not configured ▾
Enable Network Protection ⓘ	Enabled (block mode) ▾
Excluded Extensions ⓘ	

Step 4 – Assignment and Deployment

After reviewing the configuration, I assigned the antivirus policy to the **Retail_Store_Staff** group and deployed it through Microsoft Intune. Once applied, all targeted devices automatically received the Microsoft Defender security settings, ensuring a consistent security baseline across the organisation.

The screenshot displays the Microsoft Defender Antivirus configuration interface in Microsoft Intune. At the top, there are tabs for 'Basics', 'Configuration settings', 'Scope tags', 'Assignments', and 'Review + create'. The 'Assignments' tab is selected. Below the tabs is a search bar labeled 'Search by group name...'. A table lists the assigned groups:

Group	Status	Group members	Target type	Filter
Retail_Store_Staff	Active	0 devices, 4 users	Include	Add assignment filter

Below the table, the 'Create Policy' section is visible, showing the progress of the policy creation:

- Basics
- Settings: 14 settings
- Scope tags: 1 scope tags selected
- Assignments: 1 group assigned

At the bottom, two success messages are displayed in a toast notification:

- Save assignments**: Group assignments for Endpoint Antivirus Policy has been successfully saved.
- Create policy**: Endpoint Antivirus Policy has successfully been created.

Result- Microsoft Defender Antivirus policies were successfully deployed to managed devices, providing real-time threat protection, cloud-based detection capabilities, script scanning, and network protection to strengthen endpoint security across the environment.

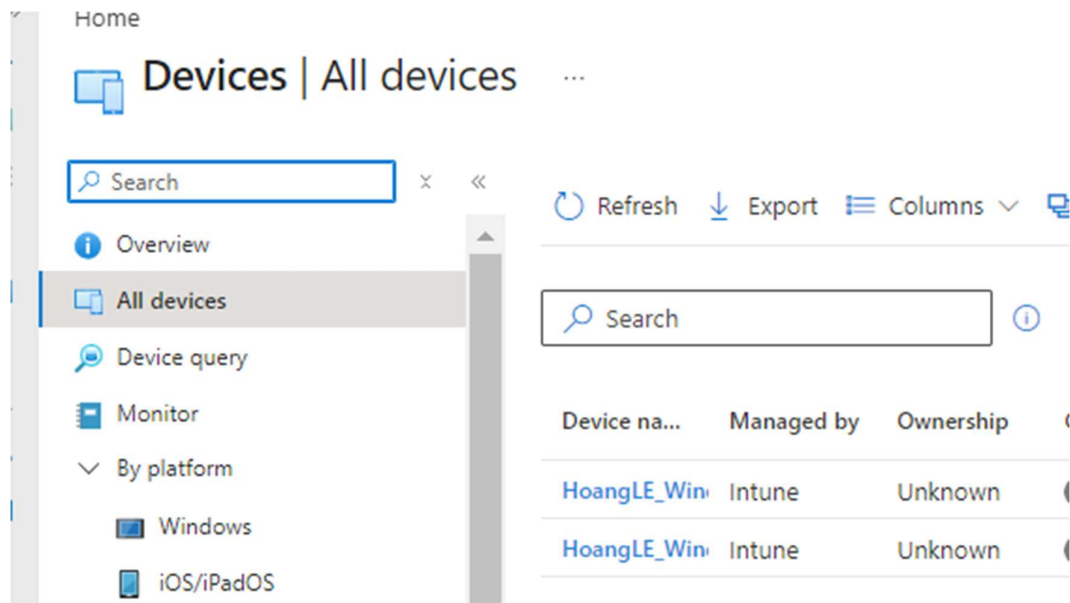
Phase 7– Test Remote Wipe / Retire Device

Objective

Use Microsoft Intune's remote device management capabilities to securely remove organisational data and management controls from devices that are retired, reassigned, or no longer authorised to access company resources.

Step 1 – Access Device Management Actions

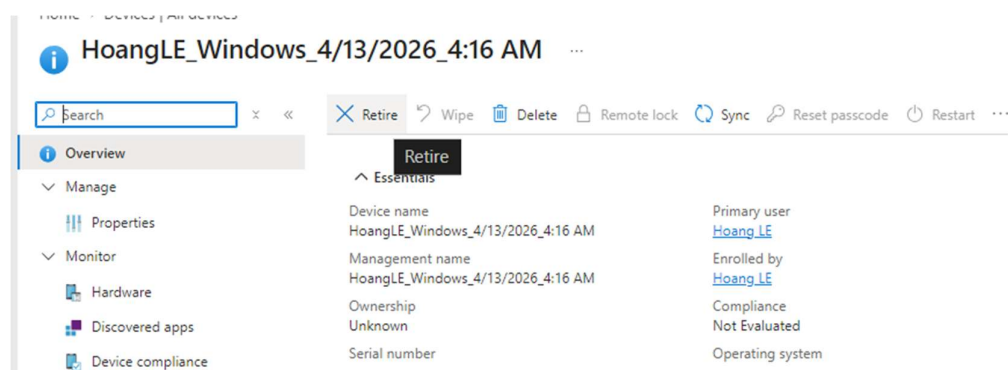
To test Microsoft Intune's remote device management capabilities, I accessed the **All Devices** section and selected a previously enrolled Windows device. From the device management page, I reviewed several available administrative actions, including 'Retire', 'Wipe', 'Delete', 'Sync', and 'Restart'.



Step 2 – Select Retire Action

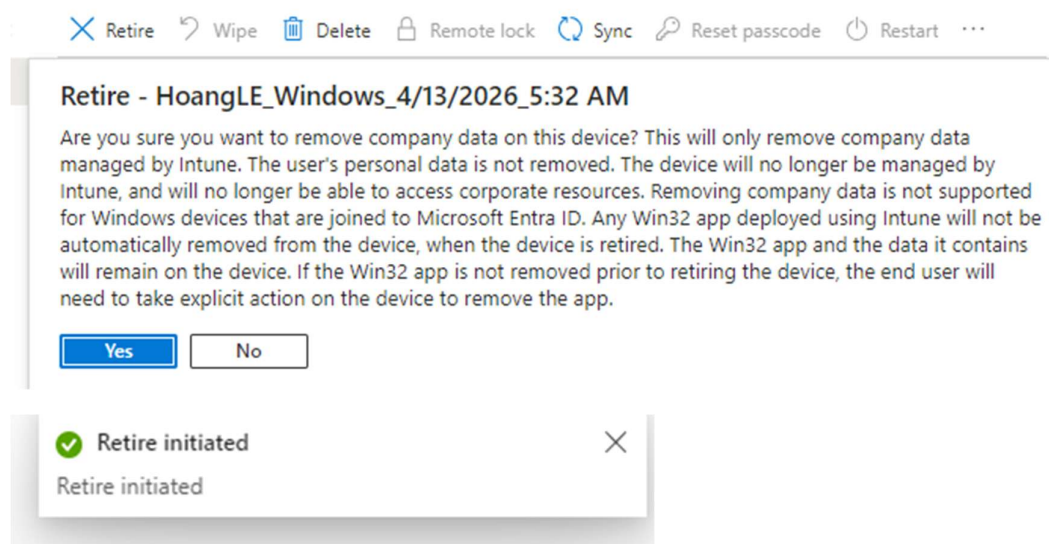
For this lab, I selected the **Retire** action to test device retirement functionality. Unlike a full device wipe, the retire process removes organisational management, policies, applications, and company data from the device while leaving the user's personal data intact.

This approach is commonly used when a device is being reassigned, replaced, or is no longer authorised to access organisational resources.



Step 3 – Confirm Retirement Process

Once I selected Retire, Intune displayed a confirmation message explaining that the action would remove company data and unmanage the device from the Intune environment. After reviewing the message, I confirmed the action and successfully initiated the retirement process.



Result- The device was successfully retired from Microsoft Intune, demonstrating how administrators can remotely remove organisational access and management controls while maintaining a secure endpoint lifecycle management process.

Phase 8 – Explore Windows Autopilot Basics

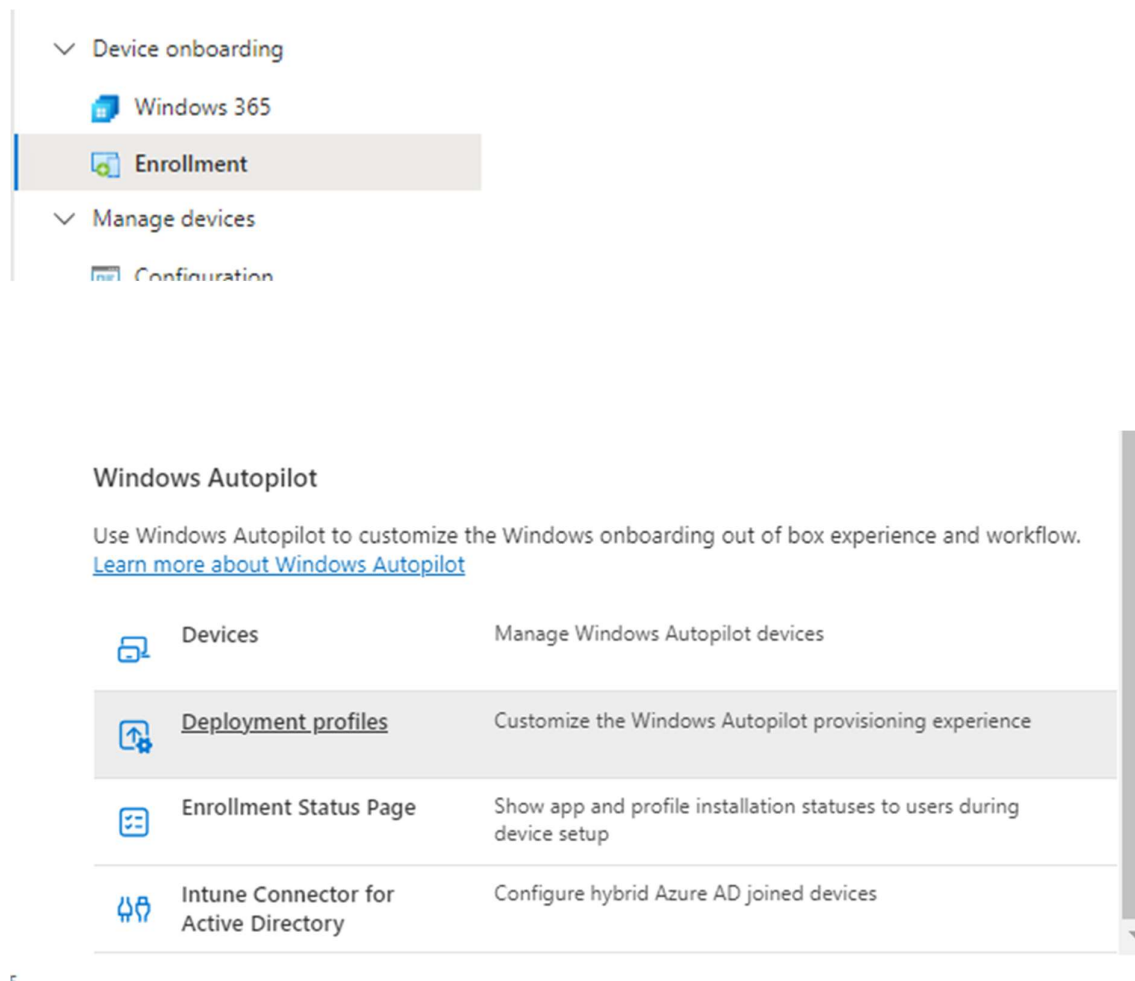
Objective

Configure a Windows Autopilot deployment profile to automate device onboarding and simplify the out-of-box experience (OOBE) for new corporate devices.

Step 1 – Access Windows Autopilot Deployment Profiles

To explore automated device provisioning, I accessed the **Windows Autopilot** section within Microsoft Intune and navigated to **Deployment Profiles**. Windows Autopilot allows organisations to preconfigure device deployment settings, reducing the amount of manual setup required when new devices are issued to users.

From there, I selected **Create Profile** and chose **Windows PC** to create a new deployment profile for managed Windows devices.



Step 2 – Create Autopilot Profile

In the profile configuration, I created a deployment profile named **Windows Autopilot Basic Profile** and provided a description to clearly identify its purpose within the Intune environment.

I also configured **Convert all targeted devices to Autopilot** as No, as this lab focused on learning and configuring Autopilot deployment settings rather than automatically registering existing production devices into the Autopilot service.

The screenshot shows the 'Windows Autopilot deployment profiles' page in the Microsoft Intune console. The left sidebar contains navigation icons for Home, Reports, Settings, Policies, Profiles, and Security. The main header shows the breadcrumb 'Home > Windows | Enrollment' and the title 'Windows Autopilot deployment profiles'. Below the header, there are buttons for '+ Create profile', 'Refresh', 'Export', and 'Columns'. A dropdown menu is open under '+ Create profile', showing 'Windows PC' and 'HoloLens'. Below the menu, there is a table with columns 'Name' and 'Description'. The table is currently empty. Below the table, there are tabs for 'Basics', 'Out-of-box experience (OOBE)', 'Assignments', and 'Review + create'. The 'Basics' tab is selected. In the 'Basics' tab, there is a 'Name' field with the value 'Windows Autopilot Basic Profile' and a green checkmark. Below it is a 'Description' field with the value 'Basic deployment profile for corporate Windows devices'. At the bottom, there is a toggle switch for 'Convert all targeted devices to Autopilot' set to 'No'. A note at the bottom states: 'By default, this profile can only be applied to Autopilot devices synced from the Autopilot service. [Learn more](#)'.

Step 3 – Configure Out-of-Box Experience (OOBE)

Next, in the **Out-of-Box Experience (OOBE)** settings, I configured **Deployment Mode** to **User-Driven**, allowing end users to complete the setup process themselves when receiving a new device. I then set **Join to Microsoft Entra ID As** to **Microsoft Entra Joined**, ensuring devices would join directly to the organisation's cloud identity environment.

✓ Basics

2 Out-of-box experience (OOBE)

3 Assignments

4 Review + create

Configure the out-of-box experience for your Autopilot devices

Deployment mode * ⓘ

User-Driven

Join to Microsoft Entra ID as * ⓘ

Microsoft Entra joined

Microsoft Software License Terms ⓘ

Show

Hide

ⓘ Important information about hiding license terms

Privacy settings ⓘ

Show

Hide

ⓘ The default value for diagnostic data collection has changed for devices running Windows 10, version 1903 and later, or Windows 11.

Hide change account options ⓘ

Show

Hide

User account type ⓘ

Administrator

Standard

Allow pre-provisioned deployment ⓘ

No

Yes

Language (Region) ⓘ

Operating system default

Automatically configure keyboard ⓘ

No

Yes

Apply device name template ⓘ

No

Yes

Create a unique name for your devices. Names must be 15 characters or less, and can contain letters (a-z, A-Z), numbers (0-9), and hyphens. Names must not contain only numbers. Names cannot include a blank space. Use the %SERIAL% macro to add a hardware-specific serial number. Alternatively, use the %RANDx% macro to add a random string of numbers, where x equals the number of digits to add.

Enter a name *

CORP-%SERIAL%

✓

Additionally, I set **Allow Pre-Provisioned Deployment** to **No**, as this deployment method was not required for the scope of this project.

Step 4 – Assignment and Deployment

After completing the deployment configuration, I moved to the **Assignments** section and assigned the Autopilot deployment profile to the **Retail_Store_Staff** group so that the deployment settings would apply to the intended users and organisational devices.

✓ Basics ✓ Out-of-box experience (OOBE) **3 Assignments** 4 Review + create

Included groups

 Add groups  Add all devices

Groups	Status	Remove
Retail_Store_Staff	Active	Remove

Excluded groups

 When excluding groups, you cannot mix user and device groups across include and exclude. [Click here to learn more about excluding groups.](#)

 Add groups

Groups	Status	Remove
No groups selected		

Finally, I reviewed all settings in the **Review + Create** section and clicked **Create** to successfully deploy the Windows Autopilot profile.

Home / Windows Autopilot deployment profiles / Windows Autopilot Basic Profile

Windows Autopilot Basic Profile | Properties

Windows PC

Search

Overview

Manage

Properties

Assigned devices

Profile saved

Basics Edit

Name	Windows Autopilot Basic Profile
Description	Basic deployment profile for corporate Windows devices
Convert all targeted devices to Autopilot	No
Device type	Windows PC

Out-of-box experience (OOBE) Edit

Deployment mode	User-Driven
Join to Microsoft Entra ID as	Microsoft Entra joined
Language (Region)	Operating system default
Automatically configure keyboard	Yes
Microsoft Software License Terms	Hide
Privacy settings	Hide
Hide change account options	Hide
User account type	Standard
Allow pre-provisioned deployment	No
Apply device name template	Yes
Enter a name	CORP-%SERIAL%

Assignments Edit

Included groups

Group	Status
Retail_Store_Staff	Active

Excluded groups

Profile successfully assigned

Go to Windows Autopilot Basic Profile to view the details.

Profile successfully created

Go to Windows Autopilot Basic Profile to view the details.

devices. [Learn more.](#)

Result- A Windows Autopilot deployment profile was successfully configured to support automated device onboarding, Microsoft Entra ID integration, and a standardised setup experience for future corporate Windows devices.

Outcome and Validation

Throughout this project, I successfully enrolled and managed Windows devices using Microsoft Intune. I implemented compliance policies, configuration profiles, application deployment, Windows Update management, endpoint security controls, device retirement actions, and Windows Autopilot configuration.

All policies were successfully created, assigned, and deployed to the Retail_Store_Staff group, demonstrating how Microsoft Intune can be used to centrally manage and secure endpoints within a cloud-based environment.

What I Learned

This project provided hands-on experience with Microsoft Intune and modern endpoint management. I learned how to enroll devices, deploy applications, enforce compliance requirements, manage Windows updates, configure Microsoft Defender security policies, perform remote device retirement actions, and explore Windows Autopilot deployment workflows.

The lab also improved my understanding of how organisations use Intune to standardise device management, automate administrative tasks, and strengthen endpoint security across large-scale environments.